Reading a Controller's Response: Step Response and Second-Order Systems

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Where this fits:

- **Builds on:** Derivative action (PD control); PID control.
- **Leads to:** Transfer functions; Resonance and bandwidth; Model-based tuning; Driving along a line; Driving to a pose (polar-coordinate controller); Hard and soft contact.

1. Overview

Key point: A step command, such as "drive at 0.5 m/s now" or "face 0.5 rad now", and the robot's response to it give four numbers: rise time, overshoot, settling time and steady-state error. When the loop's error follows a first-order or second-order equation, one or two numbers set by the gains predict all four, so we can choose gains for the response we want.

The robot of the earlier chapters plans a route (see planning with motion limits), and its feedback loops now make its speed and its **heading** (G-2284), the direction it faces, follow commands. A **proportional control** (G-2958) loop pushes in proportion to the speed error (closed-loop control), and a **PD control** (G-2962) loop, with a P part and a D part, turns the robot 45° without passing the target. But its D gain, 0.283, was found by trial: too little derivative swung the robot to 58.8°, too much made it take 10.94 s to settle. Trial finds one good pair of gains; it does not say why that pair is good, how good it is in numbers, or what to change when we want something else. Two steps the planner sends at a waypoint show the problem:

- **speed up from rest to 0.5 m/s:** the speed loop of feedback basics pushes 1 N plus 20 N per m/s of error, but from rest that gain asks for 10 N on top of the 1 N, more than the 4 N the motors give. For this test only, we drop the fixed 1 N, to see P control alone, and lower the gain from 20 to 6 N per m/s, which asks for 3 N at the start, within the limit; the speed loops of the other Notes keep 20 N per m/s. The speed then levels off at 0.375 m/s, 25 % below the command;
- **turn 0.5 rad (29°) to face the next waypoint, quicker than the 45° turn of PD control:** a first try before any tuning doubles that turn's P gain, from 0.2 to 0.4 $\mathrm{N \cdot m/rad}$, and lowers its D gain from 0.283 to 0.08 $\mathrm{N \cdot m \cdot s/rad}$, a lightly damped loop whose overshoot is easy to see and measure. The robot swings to 0.763 rad, 53 % past the target, rocks back and forth, and stays outside 2 % of the target until 9.8 s. (The turn is 0.5 rad, not 45°, because a P gain of 0.4 on 45° would ask for 0.31 $\mathrm{N \cdot m}$, more than the wheels' 0.2 $\mathrm{N \cdot m}$.)

Figure 1 animates the turn. Watch the heading pass the dashed target line, swing back below it and rock for several seconds. The gains do turn the robot, but nothing yet says in numbers how good the turn is, why it swings, or which gains would turn it quickly and cleanly.

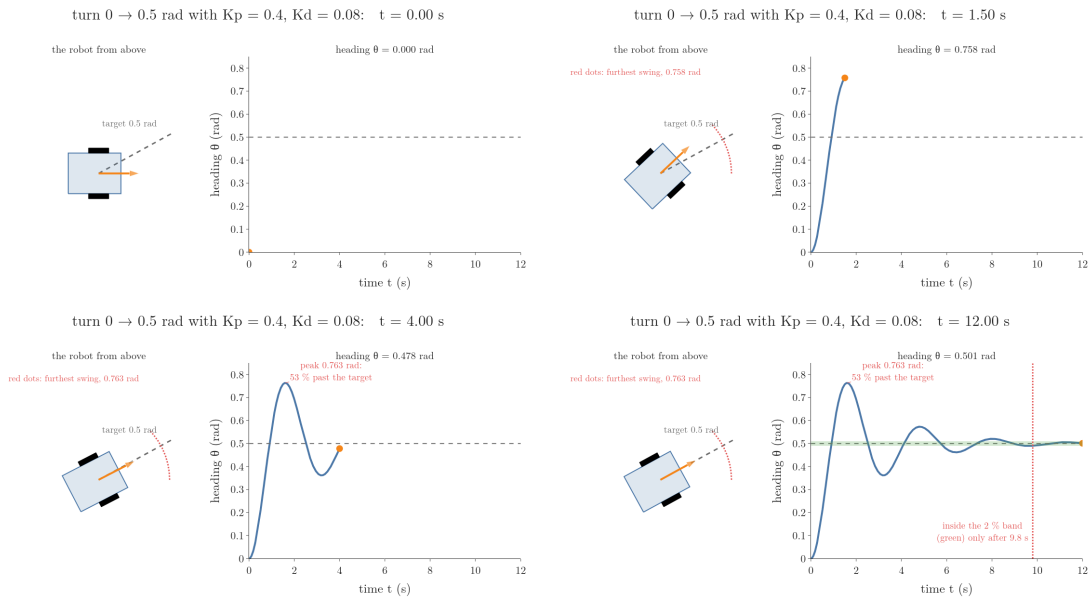


Figure 1: The opening problem: the robot turns in place from 0 to 0.5 rad with the first-try gains, 0.4 N · m/rad and 0.08 N · m · s/rad. Left: the robot from above, the target direction dashed, the red dotted arc the furthest swing so far. Right: the heading over time. The last frame shows the peak, 0.763 rad, 53 % past the target, and the green band of 2 % around the target, which the heading enters for good only after 9.8 s

This Note answers those three questions on the robot's two loops:

- the four numbers that describe any step response (Section 2);
- the error dynamics: the controller and the robot together give one equation for the error, a spring and a damper (Section 3);
- the speed loop as a first-order system: its time constant, rise time, DC gain and its steady-state error of 0.125 m/s (Section 4);
- the heading loop as a second-order system: its natural frequency, damping ratio and three kinds of response (Section 5);
- overshoot, peak time and settling time from those two numbers, and gains that turn the robot in 2.92 s without passing the target (Section 6);
- where real robot software tunes against the same numbers (Section 7).

2. The step response and its four numbers

> **Key point:** A **step input** (G-2980) jumps at once from one steady value to another, and the output that follows is the **step response** (G-2981). Four numbers describe it: **rise time** (G-2982), from 10 % to 90 % of the final value; **overshoot** (G-2983), how far past the final value it goes, in percent; **settling time** (G-2984), after which it stays within 2 %; and **steady-state error** (G-2985), the command minus the final value.

2.1 Reading the four numbers

"Rocks too long" and "falls short" from Figure 1 do not tell us whether new gains are better. We need numbers that any response can be measured by.

Both of the planner's commands jump at one instant: the speed command from 0 to 0.5 m/s, the heading command from 0 to 0.5 rad. A command that jumps at once from one steady value to another is a **step input** (G-2980), and the output that follows it is the **step response** (G-2981) (FBS §2.2). A step is the simplest test there is: it asks for one change and then waits, so everything the output does afterwards comes from the loop itself, much as tapping a bell once and listening to it ring tells us about the bell.

Figure 2 draws both responses, then reads off four numbers, one stage at a time. Watch each stage on the speed (left) and the heading (right): the two loops fail in different ways.

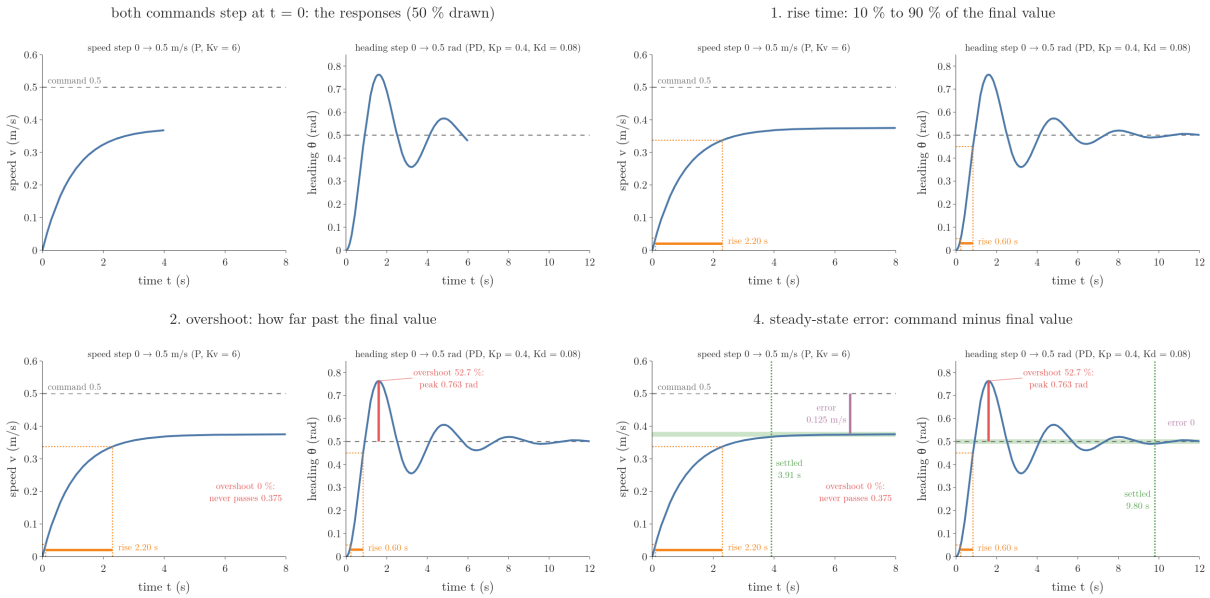


Figure 2: The four numbers read off two step responses. Left: the speed under P control with a gain of 6 N per m/s after the command steps from 0 to 0.5 m/s. Right: the heading under PD control (0.4 and 0.08) after the command steps from 0 to 0.5 rad. Stage 1, rise time: the orange bars run from 10 % to 90 % of the final value, 2.20 s and 0.60 s. Stage 2, overshoot: 0 % and 52.7 %. Stage 3, settling time: the green bands are 2 % around the final value, entered for good at 3.91 s and 9.80 s. Stage 4, steady-state error: the speed ends 0.125 m/s below its command, the heading on its target

The four numbers, each measured on Figure 2 (FBS §6.3):

1. **Rise time** (G-2982): the time to go from 10 % to 90 % of the final value; how quickly the response rises.
2. **Overshoot** (G-2983): how far the peak goes past the final value, as a percentage of the final value; how far the response swings past.

3. **Settling time** (G-2984): the time after which the response stays within 2 % of the final value; when the response has settled.
4. **Steady-state error** (G-2985): the command minus the final value; how far from the command it ends.

Books differ on the rise time of a response that swings past: we use 10 % to 90 % (FBS §6.3), while some measure from 0 % to the first time it reaches 100 % (Wikipedia, "Rise time"), which gives a different number for the same curve. The heading's overshoot, from its peak of 0.7633 rad:

$$\text{overshoot} = \frac{0.7633 - 0.5}{0.5}$$

$$= 0.5266$$

The overshoot is 52.7 percent. All four numbers of both responses:

Number	Speed (P, gain 6)	Heading (PD, 0.4 and 0.08)
rise time	2.20 s	0.60 s
overshoot	0 %	52.7 %
settling time	3.91 s	9.80 s
steady-state error	0.125 m/s	0 rad

The same numbers state what we want before any gains are chosen, as a requirement the finished loop must meet:

- **the turn:** no overshoot, so the robot never swings past the direction of a doorway, and settled within 3 s;
- **the speed:** a final speed within 2 % of the command, a steady-state error under 0.01 m/s.

Neither loop meets its requirement yet. The table describes the responses we have; it does not say why the heading swings past or which gains would stop it. Before we look for the equation behind the numbers, we note which outputs they can describe.

2.2 Not every step response looks like a step

The four numbers assume an output that rises to a level, like the speed and the heading in Figure 2. One step command also drives other outputs, and they can look quite different.

Figure 3 steps the speed command from 0 to 0.5 m/s once more and draws three outputs of the same loop. Watch how differently they end:

- **the speed** rises to a level, 0.375 m/s, and the four numbers apply;
- **the distance driven** grows like a ramp and never settles, so it has no settling time and no final value to measure overshoot from;

- **the drive force** jumps to 3 N and falls to 0.75 N, so it starts high and fades; an overshoot of a curve that starts at its peak means nothing.

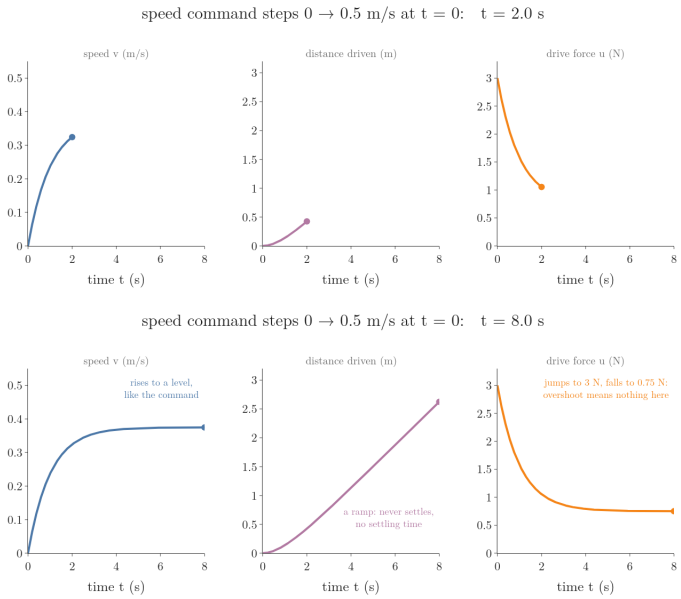


Figure 3: One step command, three step responses. The speed command steps from 0 to 0.5 m/s at t = 0 and the P speed loop responds. Left: the speed rises to a level, 0.375 m/s. Middle: the distance driven grows like a ramp and never settles. Right: the drive force jumps to 3 N and falls to 0.75 N

We therefore cannot tell from the shape alone that a curve is a step response: we must know that the input was a step and what the output is. Overshoot, rise time and settling time suit an output that should follow its command, such as the speed and the heading. An output judged by how fast it returns after a push, such as the force here or the error after a disturbance, needs other measures. The speed and the heading are outputs that follow their commands, so the four numbers apply to them; to predict those numbers from the gains, we need the equation each loop follows.

3. Error dynamics: the loop as a spring and a damper

> **Key point:** Putting the controller's rule into the robot's equation of motion gives one equation for the error alone, the **error dynamics** (G-2986). For the heading under PD control it is a block on a spring (the P gain K_p) and a damper (the D gain K_d), with the yaw inertia I as its mass, and the weak damper of the first try, 0.08 N · m · s/rad, is why the turn swings 53 % past.

Why does the heading swing 53 % past 0.5 rad, while the speed rises slowly and stops 0.125 m/s below the command? Each loop has an equation that answers it.

3.1 The heading error is a block on a spring and a damper

At the start of the turn the heading error is 0.5 rad. The P part of the controller turns the robot with a torque in proportion to the error, like a stretched spring pulling a block back towards its rest position. As the robot turns, the error shrinks, but the robot is now spinning, and only the D part, which pushes against the turn rate, can stop it at 0.5 rad. The D part works like a damper, the shock absorber of a car, which resists motion in proportion to speed. Designing the controller is like choosing the spring and the damper (MR 11.2.2). With a damper of 0.08 the robot arrives still spinning and swings past, as a block on a spring with a worn shock absorber would.

Figure 4 runs a block on a spring and a damper next to the heading error of the turn in Figure 1. The block's position is the error. Watch the block pass the dashed zero line just as the heading passes its target, and swing out to −0.263 rad: the 53 % overshoot of Figure 1, seen as an error.

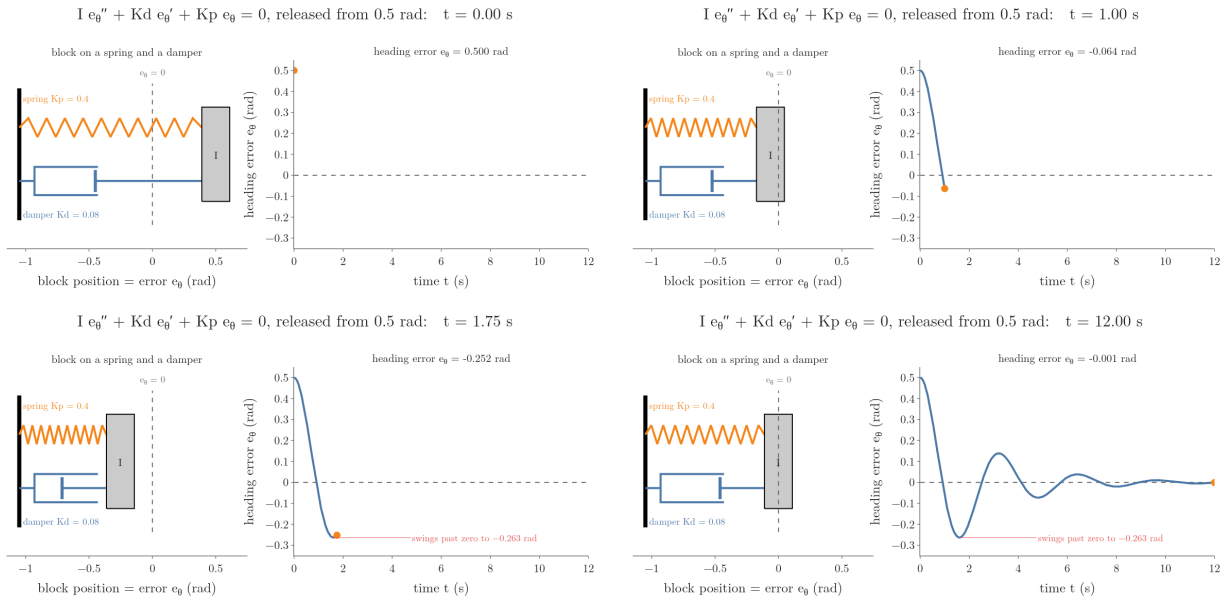


Figure 4: The heading loop as a block on a spring and a damper, run with the first-try gains. Left: a block of mass I (the yaw inertia, 0.1 kg · m^2) between a wall, a spring of stiffness 0.4 (the P gain) and a damper of 0.08 (the D gain); its position is the heading error, and the dashed line is zero error. Right: the heading error over time, released from 0.5 rad. The last frame shows the error swinging past zero to −0.263 rad at 1.60 s and dying away in smaller swings

The robot turns by **Newton's second law for rotation** (G-2957): the torque equals the **moment of inertia** (G-2955), how hard the robot is to spin, times the angular acceleration, the turning form of **Newton's second law** (G-2809). The symbols, with a dot for one **derivative** (G-595) in time and two dots for two:

- θ is the heading, in rad (0 at the start);

- $\dot{\theta}$ is the turn rate, in rad/s, measured by the gyro, and $\ddot{\theta}$ the angular acceleration, in rad/s^2;
- θ_r is the commanded heading, 0.5 rad after the step;
- e_θ is the heading error, the commanded heading minus the heading, in rad (0.5 rad at the start); the subscript keeps it apart from the number e, 2.718, of the exponential;
- I is the yaw inertia, the robot's moment of inertia about its vertical axis, 0.1 kg $\cdot$ m^2;
- τ is the turning **torque** (G-2808) the wheels apply, in N $\cdot$ m, at most 0.2 N $\cdot$ m either way;
- K_p is the P gain, the **proportional gain** (G-2959), 0.4 N $\cdot$ m/rad;
- K_d is the D gain, the **derivative gain** (G-2961), 0.08 N $\cdot$ m $\cdot$ s/rad, acting on the turn rate the gyro measures.

The robot's turning law and the PD rule, with our numbers:

$$0.1\,\ddot{\theta} = \tau$$

$$\tau = 0.4\,e_\theta - 0.08\,\dot{\theta}$$

The command is constant after the step, so the heading is 0.5 minus the error, and the heading's derivatives are minus the error's:

$$0.1\,\ddot{\theta} = 0.4\,e_\theta - 0.08\,\dot{\theta}$$

$$-0.1\,\ddot{e}_\theta = 0.4\,e_\theta + 0.08\,\dot{e}_\theta$$

$$0.1\,\ddot{e}_\theta + 0.08\,\dot{e}_\theta + 0.4\,e_\theta = 0$$

The same steps in symbols:

$$I\ddot{\theta} = K_p e_\theta - K_d\dot{\theta}$$

$$\theta = \theta_r - e_\theta$$

$$\dot{\theta} = -\dot{e}_\theta$$

$$\ddot{\theta} = -\ddot{e}_\theta$$

$$-I\ddot{e}_\theta = K_p e_\theta + K_d\dot{e}_\theta$$

$$I\ddot{e}_\theta + K_d\dot{e}_\theta + K_p e_\theta = 0$$

A block on a spring and a damper follows the same equation: its mass times its acceleration, plus the damper's constant times its speed, plus the spring's stiffness times its position, is zero (MR 11.2.2). Here the yaw inertia is the mass, K_d is the damper and K_p is the spring. An equation that says how the error of a controlled system changes over time is its **error dynamics** (G-2986) (MR 11.2.1). The heading error starts at 0.5 rad, with the robot at rest, so with no turn rate. The robot's controller runs 100 times a second; we treat it as acting continuously, which is close because 0.01 s is short next to the swings here, and digital control treats the difference.

3.2 The speed error has no mass to swing

The speed loop gives its own error equation. The symbols:

- v is the speed, in m/s (0 at the start), and $\dot{v}$ its rate of change, in m/s^2;
- v_r is the commanded speed, 0.5 m/s after the step;
- e_v is the speed error, the commanded speed minus the speed, in m/s (0.5 m/s at the start);
- m is the robot's mass, 8 kg;
- b is the rolling drag, 2 N $\cdot$ s/m: at a speed of 1 m/s the floor holds the robot back with 2 N;
- u is the drive force, in N, at most 4 N either way;
- K_v is the speed loop's P gain, 6 N per m/s.

Newton's second law with the drag, the P rule put in for the force, then the speed written as 0.5 minus the error:

$$8\,\dot{v} = 6\,e_v - 2\,v$$

$$-8\,\dot{e}_v = 6\,e_v - 2\,(0.5 - e_v)$$

$$-8\,\dot{e}_v = 8\,e_v - 1$$

$$8\,\dot{e}_v + 8\,e_v = 1$$

In symbols:

$$m\dot{v} = K_v e_v - b\,v$$

$$-m\dot{e}_v = K_v e_v - b\,(v_r - e_v)$$

$$m\dot{e}_v + (K_v + b)\, e_v = b\, v_r$$

Two things differ from the heading error. Only the first derivative appears: the robot's mass is pushed by the force, but no term stores a swing the way the heading's spring does. And the right-hand side is not zero but 1 N, the drag at the commanded speed, which the loop must keep supplying (MR 11.4 meets the same kind of constant, the pull of gravity on an arm). The feedback basics loop adds a fixed 1 N push, found by a test on the flat floor, which balances that drag there; we left it out to see what P control alone does, and without it the error shows plainly. With it, the error comes back wherever the test does not hold, such as on the ramp. The heading error follows a second-order equation and the speed error a first-order one; Section 4 solves the first.

4. The speed loop: a first-order system

> **Key point:** A **first-order system** (G-2987) closes the same fraction of its remaining gap in every **time constant** (G-2988) T, 63 % after T and 98 % after $4T$; the speed loop's T of 1 s gives a rise of 2.20 s and a settling time of 3.91 s. The speed ends at the **DC gain** (G-2989), 0.75, times the command, 0.375 m/s, because a P loop needs an error to keep pushing against the drag.

4.1 The time constant

The speed error of Section 3.2 obeys an equation with one derivative. What does its solution look like, and how fast does it close the gap?

Divide the equation by 8: the error changes at a rate equal to minus its distance from 0.125 m/s. A gap that shrinks at a rate in proportion to its own size shrinks by the same fraction in every second, as a cup of hot tea loses the same fraction of its extra heat every minute. The solution, with t the time in s since the step:

$$e_v = 0.125 + 0.375\, e^{-t}$$

$$v = 0.375\, (1 - e^{-t})$$

Why it holds. At the start, $e^0 = 1$, so the error is 0.5 m/s, as it should be. By the chain rule, the derivative of e^{-t} is $-e^{-t}$. Put both into the left-hand side of Section 3.2's equation:

$$8\, \dot{e}_v = -3\, e^{-t}$$

$$8\, e_v = 1 + 3\, e^{-t}$$

$$8\,\dot{e}_v + 8\,e_v = 1$$

which is the right-hand side. A system whose output follows a first-order differential equation, one with only a first derivative, is a **first-order system** (G-2987) (Chen L6). The time it takes to close 63 % of the gap is its **time constant** (G-2988) T. The symbols:

- T is the time constant, in s;
- v_∞ is the final speed, in m/s, 0.375 m/s here.

Dividing Section 3.2's equation by $K_v + b$ leaves 1 in front of the error, and the time constant is what remains in front of its derivative:

$$T = \frac{m}{K_v + b}$$

$$= \frac{8}{6 + 2} = 1 \text{ s}$$

$$v = v_\infty\left(1 - e^{-t/T}\right)$$

Many books write the time constant as τ (Chen L6); in this Note τ is the turning torque, so the time constant is T, as in FBS. In each time constant the gap shrinks to e^{-1}, 37 %, of what it was, so the speed has closed (Chen L6):

Time	Fraction of the way to 0.375 m/s
$T = 1$ s	63.2 %
$2T = 2$ s	86.5 %
$3T = 3$ s	95.0 %
$4T = 4$ s	98.2 %

Figure 5 shows this in two stages. In stage 1, watch the orange dots appear at 1, 2, 3 and 4 s on the speed curve, each closing about two thirds of what is left. The purple dashed line is the starting slope: had the speed kept its first rate of rise, 0.375 m/s^2, it would have reached 0.375 m/s in exactly one time constant (Chen L6).

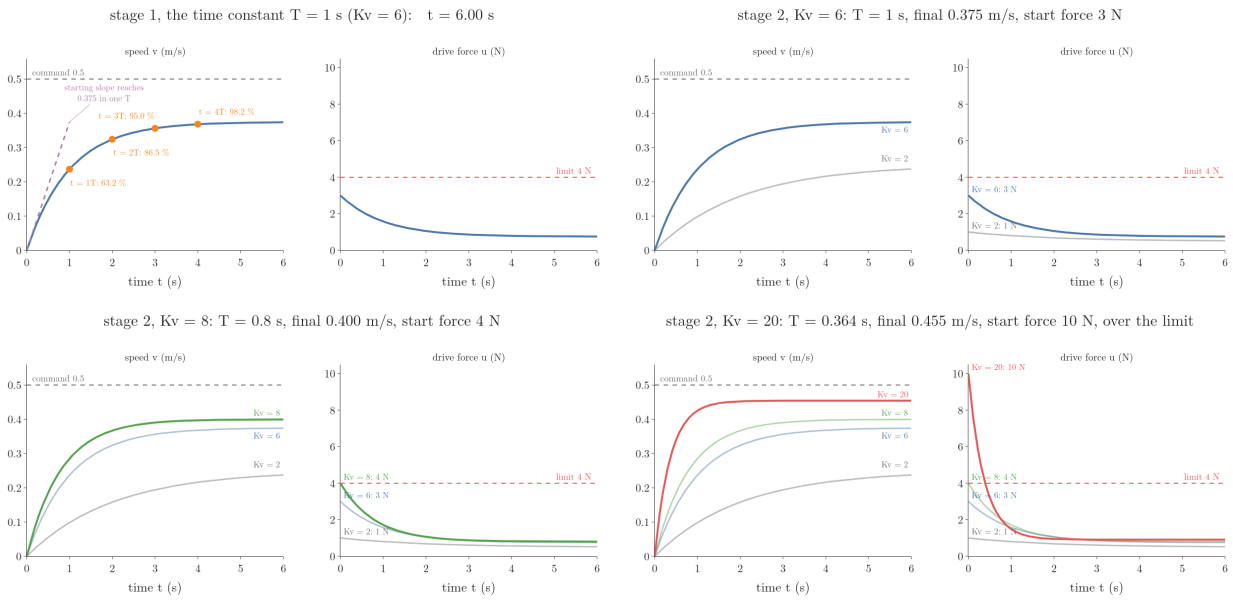


Figure 5: The speed loop as a first-order system. Stage 1: the speed under P control (gain 6, time constant T = 1 s) is drawn over time; orange dots at 1, 2, 3 and 4 s mark 63.2, 86.5, 95.0 and 98.2 % of the way to 0.375 m/s; the purple dashed line is the starting slope, which reaches 0.375 m/s in one time constant. Right: the drive force falls from 3 N. Stage 2: the gain is raised to 2, 6, 8 and 20 one after another; a larger gain gives a shorter time constant and a final speed closer to 0.5 m/s, but the gain of 20 asks for 10 N at the start, over the 4 N limit (red dashed line)

4.2 Rise time and settling time from the time constant

The time constant fixes the other numbers of Section 2. The speed reaches 10 % of its final value when the exponential has fallen to 0.9, and 90 % when it has fallen to 0.1. With $T = 1$ s:

$$1 - e^{-t_1} = 0.1$$

$$t_1 = \ln(1/0.9) = 0.1054 \text{ s}$$

$$1 - e^{-t_2} = 0.9$$

$$t_2 = \ln 10 = 2.3026 \text{ s}$$

$$T_r = 2.3026 - 0.1054 = 2.1972 \text{ s}$$

In symbols, with T_r the rise time:

$$T_r = T \ln 10 - T \ln(1/0.9)$$

$$= T \ln 9 = 2.197\, T$$

The speed is within 2 % of its final value once the exponential has fallen to 0.02. With T_s the settling time:

$$e^{-T_s/T} = 0.02$$

$$T_s = T \ln 50 = 3.912\, T$$

With $T = 1$ s these are the 2.20 s and 3.91 s measured in Figure 2; the usual rules of thumb are $2.2T$ and $4T$ (Chen L6; MR 11.2.2.1). One number, the time constant, gives both, and it falls when the gain rises.

4.3 DC gain and steady-state error

Why does the speed stop at 0.375 m/s? Once the speed stops changing, its rate of change is zero, and the force left over only balances the drag:

$$0 = 6\,(0.5 - v_\infty) - 2\, v_\infty$$

$$8\, v_\infty = 3$$

$$v_\infty = 0.375 \text{ m/s}$$

In symbols, the final speed is a fixed fraction of the command:

$$v_\infty = \frac{K_v}{K_v + b}\, v_r$$

$$e_{ss} = v_r - v_\infty$$

$$= \frac{b}{K_v + b}\, v_r$$

with e_{ss} the steady-state error, 0.125 m/s. The ratio of a stable system's final output to the size of its step is its **DC gain** (G-2989) (Chen L6). Here:

$$\frac{K_v}{K_v + b} = \frac{6}{8} = 0.75$$

so the robot reaches three quarters of the commanded speed.

The mechanism: to hold 0.375 m/s the drive must keep pushing against the drag with:

$$2 \times 0.375 = 0.75 \text{ N}$$

A P loop pushes only in proportion to the error, so it needs an error that gives 0.75 N:

$$6 \times e_{ss} = 0.75$$

$$e_{ss} = 0.125 \text{ m/s}$$

The error is what sustains the push (MR 11.4).

4.4 A larger gain: faster and closer, but more force

Can a larger gain remove the 0.125 m/s error? Stage 2 of Figure 5 raises the gain from 2 to 20, the gain of feedback basics. Watch the speed curves rise faster and end closer to 0.5 m/s, and the force curves on the right start higher. The numbers, from the formulas above (notebook):

P gain (N per m/s)	Time constant	Final speed	Steady-state error	Force at the start
2	2 s	0.25 m/s	0.25 m/s	1 N
6	1 s	0.375 m/s	0.125 m/s	3 N
8	0.8 s	0.4 m/s	0.1 m/s	4 N
20	0.364 s	0.455 m/s	0.045 m/s	10 N

The requirement of Section 2.1, an error under 0.01 m/s, needs:

$$\frac{2 \times 0.5}{K_v + 2} = 0.01$$

$$K_v = 98 \text{ N per m/s}$$

At the start that gain asks for a force of:

$$98 \times 0.5 = 49 \text{ N}$$

twelve times the 4 N the motors can give. Within the limit, the smallest error P control can reach is 0.1 m/s, with a gain of 8. **Integral action** (G-2963), a push in proportion to the error's running sum, supplies the 0.75 N without needing an error, which is why it exists, but it adds a second state, the summed error, and so makes the speed loop second order. The first-order loop cannot overshoot, because its single exponential only approaches the final value. The heading loop's second-order error can swing past, and it needs two numbers.

5. The heading loop: a second-order system

> **Key point:** Divided by the yaw inertia, the heading error equation takes the standard form of a **second-order system** (G-2990), set by two numbers: the **natural frequency** (G-2991) ω_n, 2 rad/s, which sets how fast the error swings, and the **damping ratio** (G-2992) ζ, 0.2, which sets how much the swing is damped. Below 1 the response overshoots (**underdamped**, G-2994); at 1 it is the fastest response with no overshoot (**critically damped**, G-2995); above 1 it rises slowly (**overdamped**, G-2996).

5.1 Natural frequency and damping ratio

Section 3.1 gave the heading error equation for the first-try gains, 0.4 and 0.08. Which combination of its three numbers makes the turn swing 53 % past?

Two plain questions decide it. How fast would the robot swing if there were no damper at all? And how strong is the damper, compared with the damper that would just stop the swinging? Divide the equation by the yaw inertia, 0.1:

$$\ddot{e}_\theta + 0.8\,\dot{e}_\theta + 4\,e_\theta = 0$$

The standard way to write a second-order equation names those two answers (MR 11.2.2.2; Chen L7):

- ω_n is the **natural frequency** (G-2991), in rad/s: how fast the error would swing with no damping;
- ζ (zeta) is the **damping ratio** (G-2992), a number with no unit: the damping compared with the damping that just stops the swing.

$$\ddot{e}_\theta + 2\zeta\omega_n\dot{e}_\theta + \omega_n^2 e_\theta = 0$$

Matching the two equations term by term:

$$\omega_n^2 = 4$$

$$\omega_n = 2 \text{ rad/s}$$

$$2\zeta\omega_n = 0.8$$

$$\zeta = \frac{0.8}{2 \times 2} = 0.2$$

In symbols, from Section 3.1's equation divided by I:

$$\omega_n^2 = \frac{K_p}{I}$$

$$2\zeta\omega_n = \frac{K_d}{I}$$

$$\zeta = \frac{K_d}{2I\omega_n}$$

$$= \frac{K_d}{2\sqrt{K_p I}}$$

A check of the meaning. The damper that would make the damping ratio exactly 1 is:

$$2\sqrt{K_p I} = 2\sqrt{0.4 \times 0.1}$$

$$= 0.4 \text{ N} \cdot \text{m} \cdot \text{s/rad}$$

Our D gain, 0.08, is a fifth of it, which is why the damping ratio is 0.2. The same check explains the D gain that PD control found by trial, 0.283 with a P gain of 0.2. The damper that just stops the swing is then:

$$2\sqrt{0.2 \times 0.1} = 0.283$$

so that D gain gives a damping ratio of 1, which is why the swing past 45° disappeared near 0.28 there. A system whose output follows a differential equation with a second derivative, like this one, is a **second-order system** (G-2990) (Chen L6). With no damper at all, the error would swing for ever at 2 rad/s, one swing every 3.14 s; that rate is why ω_n is called the natural frequency.

5.2 Two roots decide the kind of response

How does a damping ratio of 0.2 turn into a swing past the target? We look for solutions of the form e^{st}, an exponential with some number s in its exponent: its first derivative is s times itself and its second s^2 times itself. Put in the equation of Section 5.1, every term carries e^{st}, which is never zero, so:

$$s^2 + 0.8\,s + 4 = 0$$

The equation is the **characteristic equation** (G-2993) of the error dynamics (MR 11.2.2). Its two roots, by the quadratic formula:

$$s = \frac{-0.8 \pm \sqrt{0.64 - 16}}{2}$$

$$= -0.4 \pm \frac{\sqrt{-15.36}}{2}$$

$$= -0.4 \pm 1.96\,j$$

The square root of a negative number is a complex number: j is the number whose square is -1, so the square root of -15.36 is $3.92\ j$. By Euler's formula, $e^{j\omega t}$ is $\cos\omega t + j\sin\omega t$, so a root with an imaginary part makes the error swing. Each root gives an error term made of two parts:

- the real part, -0.4, gives a fading envelope $e^{-0.4t}$: how fast the swings die;
- the imaginary part, 1.96, gives a swing at 1.96 rad/s, the **damped natural frequency** (G-2997) ω_d, slower than the natural frequency because the damper drags on every swing.

In symbols (MR 11.2.2.2; Chen L6):

$$s = -\zeta\omega_n \pm \omega_n\sqrt{\zeta^2 - 1}$$

$$\omega_d = \omega_n\sqrt{1 - \zeta^2}$$

$$= 2\sqrt{1 - 0.04} = 1.96 \text{ rad/s}$$

The sign of what is under the square root, $\zeta^2 - 1$, sorts every second-order response into one of three kinds (MR 11.2.2.2; Chen L6):

- **underdamped** (G-2994), ζ below 1: two complex roots; the error swings inside a fading envelope and the response overshoots, 52.7 % at $\zeta = 0.2$;

- **critically damped** (G-2995), $\zeta = 1$: one root, -2, twice; the fastest response that does not overshoot, settled at 2.92 s;
- **overdamped** (G-2996), ζ above 1: two different negative real roots; the response rises slowly, held back by the slower root: at $\zeta = 2$ the roots are -0.54 and -7.46 and the turn takes 7.44 s.

At $\zeta = 0$ the roots are $\pm 2j$, with no real part, and the swing never fades. Critical damping is $\zeta = 1$; a damping ratio of 0.7 is still underdamped and overshoots by 4.6 %.

Figure 6 lowers the damping ratio from 2 to 0 while the natural frequency stays at 2 rad/s, which means lowering the D gain from 0.8 to 0. Watch the two roots slide towards each other along the real axis, meet at -2 when the damping ratio reaches 1, then split up and down around the dotted circle of radius 2. At the same moment the response on the right changes from a slow rise to an overshoot.

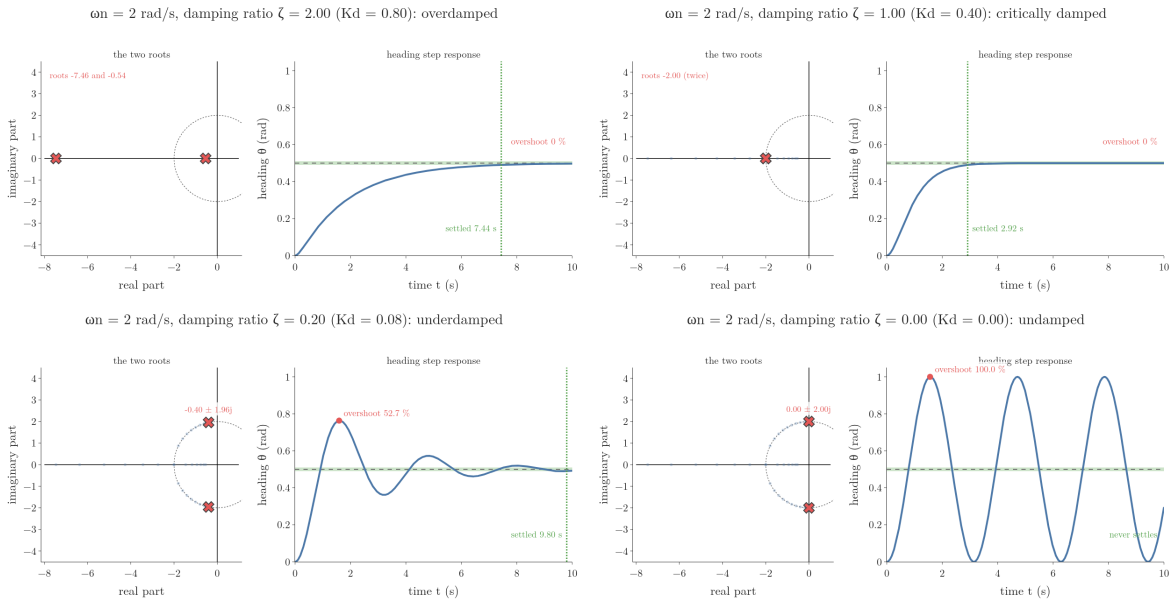


Figure 6: The damping ratio sets the kind of response. The natural frequency stays at 2 rad/s (P gain 0.4) while the damping ratio falls from 2 to 0 (D gain 0.4 times the damping ratio). Left: the two roots in the plane of complex numbers, real part across and imaginary part up, earlier positions faint, the circle of radius 2 dotted. Right: the heading step response with its overshoot and its 2 % settling time. Overdamped at 2: roots -7.46 and -0.54, settled at 7.44 s. Critically damped at 1: both roots at -2, no overshoot, settled at 2.92 s. Underdamped at 0.2: roots -0.40 ± 1.96j, overshoot 52.7 %, settled at 9.80 s. Undamped at 0: roots ± 2j, a swing that never settles

The damping ratio says which kind of response we get, and the natural frequency how fast. Section 6 turns the pair into the four numbers of Section 2, and back into gains.

6. From the two numbers to the four, and back to gains

> **Key point:** For an underdamped second-order system of the standard form, the overshoot depends only on the damping ratio, the first peak comes at the **peak time** (G-2998) π/ω_d, and the response settles in about $4/(\zeta\omega_n)$. The turn's requirement, no overshoot and settled within 3 s, gives a damping ratio of 1 and a natural frequency of 2 rad/s, so this Note's faster design is $K_p = 0.4$ and $K_d = 0.4$: the robot turns 0.5 rad in 2.92 s without passing the target.

6.1 Overshoot depends only on the damping ratio

Section 2 measured 52.7 % overshoot and 9.80 s settling on the first-try turn. Can we predict both from a damping ratio of 0.2 and a natural frequency of 2 rad/s?

The heading error of an underdamped turn that starts at rest, with ζ and ω_n from Section 5 (FBS §7.3, eq. 7.24; Chen L7):

$$e_\theta = 0.5\, e^{-\zeta\omega_n t}\, C$$

$$C = \cos\omega_d t + 0.2041\, \sin\omega_d t$$

where 0.2041 is $\zeta/\sqrt{1-\zeta^2}$ for $\zeta = 0.2$. **Why it holds** at the start: at $t = 0$ the cosine is 1 and the sine is 0, so the error is 0.5 rad. Its rate of change at $t = 0$ is 0.5 times the envelope's rate, -0.4, plus 0.5 times the sine's rate, 1.96 times 0.2041:

$$0.5\,(-0.4 + 1.9596 \times 0.2041)$$

$$= 0.5\,(-0.4 + 0.4) = 0$$

so the robot starts at rest, as it does. The heading is furthest past the target when the swing has done half a cycle (FBS §7.3). Half a cycle at 1.96 rad/s takes the **peak time** (G-2998):

$$T_p = \frac{\pi}{\omega_d} = \frac{3.1416}{1.9596}$$

$$= 1.6032 \text{ s}$$

There the cosine is -1 and the sine 0:

$$e_\theta = -0.5\, e^{-0.4\times 1.6032}$$

$$= -0.5\, e^{-0.6413}$$

$$= -0.5 \times 0.5266 = -0.2633 \text{ rad}$$

The error is 0.2633 rad past zero: an overshoot of 0.5266, the 52.7 % measured in Figure 2. In symbols, the ratio of the swing past to the step is the envelope at the peak time:

$$\text{overshoot} = e^{-\zeta\omega_n T_p}$$

$$= e^{-\zeta\omega_n \pi/\omega_d}$$

$$= e^{-\pi\zeta/\sqrt{1-\zeta^2}}$$

because ω_n/ω_d is $1/\sqrt{1-\zeta^2}$. The natural frequency has cancelled: the overshoot depends only on the damping ratio (Chen L7; FBS §7.3, Table 7.1). Read backwards, a requirement on overshoot is a requirement on the damping ratio: 5 % overshoot needs a damping ratio of 0.69 (Chen L7, which inverts the formula; notebook).

6.2 Settling time is about four envelope time constants

The swing fades inside the envelope $e^{-\zeta\omega_n t}$, a first-order fade like the speed of Section 4 with time constant $1/(\zeta\omega_n)$, 2.5 s here. By Section 4.2, a first-order fade is within 2 % after about four time constants (Chen L7; FBS Table 7.1):

$$T_s \approx \frac{4}{\zeta\omega_n} = \frac{4}{0.2 \times 2}$$

$$= 10 \text{ s}$$

The measured settling time is 9.80 s: the rule is close for underdamped responses, because they settle as their envelope does. The rule is not close for critical damping. There the error is $0.5\,(1+\omega_n t)\, e^{-\omega_n t}$ (FBS eq. 7.24), and the extra factor $1+\omega_n t$ holds it up: the rule says 2.0 s, but the response settles at 2.92 s. Check at that time, where $\omega_n t$ is 5.834:

$$(1 + 5.834) \times e^{-5.834}$$

$$= 6.834 \times 0.002926 = 0.0200$$

which is the 2 % edge (notebook).

6.3 The natural frequency only stretches time

If the damping ratio alone sets the overshoot, what does the natural frequency do? Figure 7 keeps the damping ratio at 0.2 and raises the natural frequency from 1 to 4 rad/s. Watch the roots move straight out along one ray from the origin, and the response keep its shape, overshoot 52.7 % every time, while it is squeezed in time: the settling time falls from 9.80 s at 2 rad/s to 4.90 s at 4 rad/s.

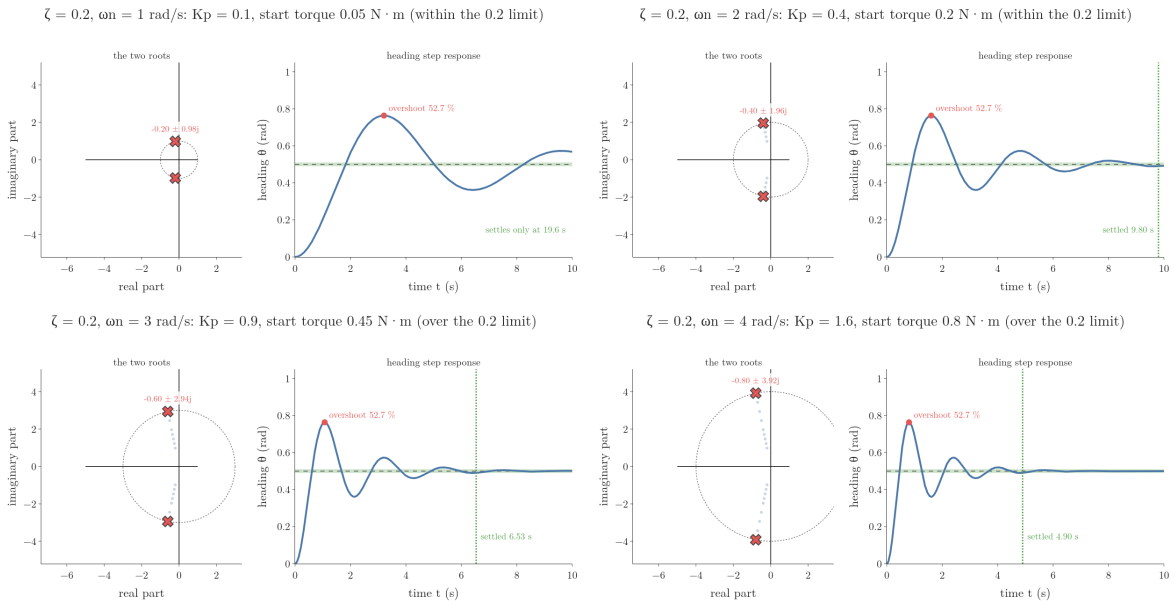


Figure 7: The natural frequency only stretches time. The damping ratio stays at 0.2 while the natural frequency rises from 1 to 4 rad/s. Left: the two roots move out along one ray from the origin, on a dotted circle whose radius is the natural frequency. Right: the heading step response keeps 52.7 % overshoot and is squeezed in time; it settles at 9.80 s at 2 rad/s and 4.90 s at 4 rad/s. The title gives the P gain, the yaw inertia times the natural frequency squared, and the torque it asks for at the start of a 0.5 rad turn: within the 0.2 N · m limit up to 2 rad/s, over it above

Time enters the response only as the product $\omega_n t$, so doubling the natural frequency halves every time in it: rise, peak and settling (FBS §7.3). The cost is torque. The natural frequency squared is K_p/I, and the P gain asks for K_p times 0.5 rad at the start of the turn. At 2 rad/s:

$$K_p = I\omega_n^2 = 0.1 \times 4 = 0.4$$

$$0.4 \times 0.5 = 0.2 \text{ N} \cdot \text{m}$$

The starting torque is exactly the 0.2 N · m the wheels can give, so 2 rad/s is the fastest natural frequency this robot can use for a 0.5 rad turn without its torque exceeding the limit.

Limits on the actuators bound the gains (MR 11.4).

6.4 Choosing the gains for the turn

The turn's requirement from Section 2.1 is no overshoot and settled within 3 s, within the 0.2 N · m torque limit. Each part fixes one number:

- **no overshoot:** a damping ratio of at least 1, and the fastest such response is critical damping, $\zeta = 1$ (MR 11.2.2.2: critical damping is a good goal);
- **settled within 3 s, within the torque:** the largest natural frequency the torque allows, 2 rad/s (Section 6.3).

The gains follow from Section 5.1's formulas:

$$K_p = I\omega_n^2 = 0.1 \times 4$$

$$= 0.4 \text{ N} \cdot \text{m/rad}$$

$$K_d = 2\zeta\omega_n I = 2 \times 1 \times 2 \times 0.1$$

$$= 0.4 \text{ N} \cdot \text{m} \cdot \text{s/rad}$$

The new gains keep the damping ratio of the tuned PD gains, 1, with twice their P gain: a natural frequency of 2 rad/s instead of 1.41 rad/s, so by Section 6.3 every time in the response is 1.41 times shorter. The tuned pair, 0.2 and 0.283, also turns cleanly but settles at 4.13 s, outside the 3 s asked for, which is why this Note designs a stiffer pair (notebook). With these gains the roots are -2 twice, the rise time is 1.68 s, there is no overshoot, and the turn settles at 2.92 s, inside the 3 s asked for. The torque starts at 0.2 N · m and never goes past it; on the way in it brakes with at most 0.027 N · m (notebook). Figure 8 runs both pairs of gains side by side. Watch the first try swing past the dashed target and rock, while the designed robot slows as it nears the target and stops on it.

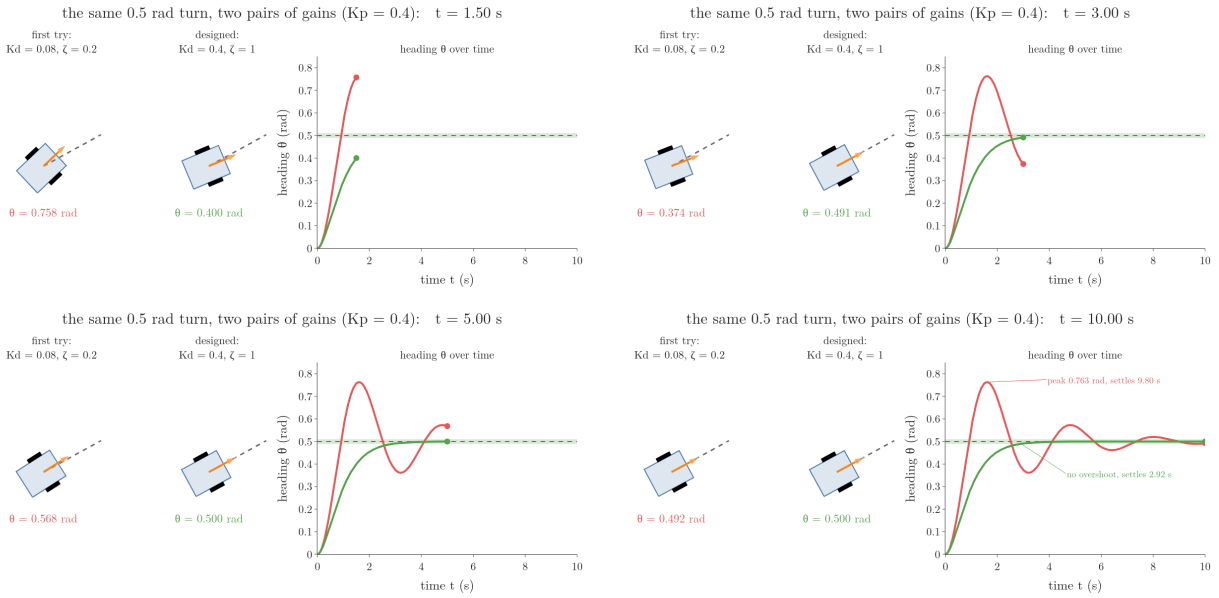


Figure 8: The fix: the same 0.5 rad turn with the first-try D gain, 0.08 (damping ratio 0.2), and the designed one, 0.4 (damping ratio 1), both with a P gain of 0.4. Left and middle: the two robots from above, target dashed. Right: both headings over time with the 2 % band. The last frame shows the first try peaking at 0.763 rad and settling at 9.80 s, and the designed turn never passing 0.5 rad and settling at 2.92 s

The choice trades speed against overshoot. If 5 % overshoot were allowed, a damping ratio of 0.7 (a D gain of 0.28) would rise in 1.06 s instead of 1.68 s and settle at 2.99 s, still within 3 s, but the robot would swing 4.6 % past the doorway's direction (notebook).

Which gains the rest of the chapter uses. The pair 0.4 and 0.4 is this Note's faster design, made for the 3 s turn, and transfer functions, poles and zeros keeps it. The other Notes of the chapter keep the tuned gains of PD and PID control: a P gain of 0.2 N · m/rad, a D gain of 0.283 N · m · s/rad and an integral gain of 0.05 N · m per rad · s. Frequency response, stability margins and sensitivity also study the first try, 0.4 and 0.08, because its large swing shows their effects plainly.

The formulas of this section hold for the standard second-order form only, with nothing else in the loop (FBS §7.3). A loop with a third state, such as an integral term or a lagging sensor, or with a term that acts on the rate of the command, responds differently. Reading such loops needs transfer functions, poles and zeros. The turn now meets its requirement: 2.92 s and no overshoot, where the first try took 9.80 s and swung 53 % past.

7. Where real stacks use these numbers

Real robot software is tuned against the same four numbers. Two places where this shows in code we read:

- **PX4's attitude autotuner** (drones). A drone's turn rate is a loop like our

heading loop, which is why it is relevant here. The tuner shakes the drone with a train of small steps on each axis, fits a second-order model of each turn rate, and computes PID gains for a requested closed-loop rise time, `MC_AT_RISE_TIME`, 0.14 s by default and 0.2 s for yaw, with a damping index of 0, which the design code documents as "critical damping" (PX4's `mc_autotune_attitude_control.cpp`, `mc_autotune_attitude_control_params.yaml` and `pid_design.hpp`; full paths in Sources). Figure 9 animates the four steps, one at a time; watch step 3, where the requirement is written in the step-response numbers of Section 2.

- **Autoware's speed controller** (cars). Its longitudinal PID controller goes into an emergency state if the car overshoots the point where it should stop by more than 1.5 m (`emergency_state_overshoot_stop_dist: 1.5` in its parameter file; the test is in `pid_longitudinal_controller.cpp`; full paths in Sources).

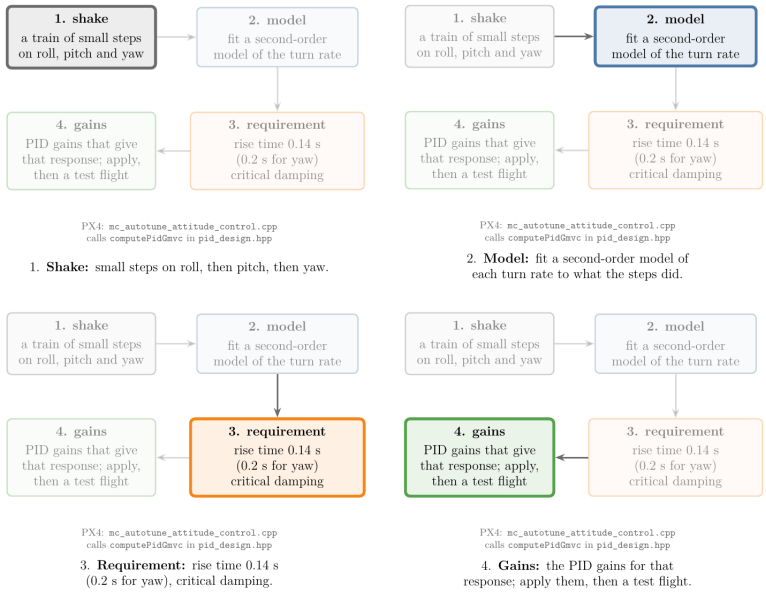


Figure 9: PX4's attitude autotuner, one step per frame, the step in use drawn thick: 1, shake the drone with a train of small steps on roll, then pitch, then yaw; 2, fit a second-order model of each turn rate; 3, the requirement: a rise time of 0.14 s (0.2 s for yaw) and critical damping; 4, the PID gains that give that response, applied and checked in a test flight. The last frame shows all four. The code: mc_autotune_attitude_control.cpp calls computePidGmvc in pid_design.hpp

8. Summary

The symbols: T the time constant, K_v the speed loop's P gain, m the mass, 8 kg, b the drag, 2 N · s/m; ω_n the natural frequency, ζ the damping ratio, ω_d the damped natural frequency, K_p and K_d the heading loop's gains, I the yaw inertia, 0.1 kg · m^2.

Idea	Formula or value	Why it matters
step response	rise, overshoot, settling, steady-state error	one test, four comparable numbers
first-order time constant	$T = m/(K_v + b) = 1$ s	sets rise ($2.2T$) and settling ($4T$)
DC gain	$K_v/(K_v + b) = 0.75$	the final speed is 0.375 m/s, not 0.5
natural frequency	$\omega_n = \sqrt{K_p/I} = 2$ rad/s	sets how fast; bounded by torque
damping ratio	$\zeta = K_d/(2\sqrt{K_p I})$	sets the kind of response and the overshoot
overshoot	$e^{-\pi\zeta/\sqrt{1-\zeta^2}}$	52.7 % at 0.2, 0 % at 1 or above
peak time	$T_p = \pi/\omega_d = 1.60$ s	when the swing past is largest
settling time	about $4/(\zeta\omega_n)$	9.80 s at 0.2; 2.92 s exactly at 1

- The first-try turn swung 53 % past and settled only at 9.80 s because its D gain of 0.08 gave a damping ratio of 0.2. With this Note's faster design, $K_p = 0.4$ and $K_d = 0.4$, a damping ratio of 1 and a natural frequency of 2 rad/s, the robot turns 0.5 rad in 2.92 s without passing the target, at the 0.2 N · m torque limit, so the turn meets its requirement; the rest of the chapter keeps the tuned gains of PD control, 0.2 and 0.283, which settle at 4.13 s.
- The speed stopped at 0.375 m/s, 0.125 m/s below its command, because a P loop needs an error of 0.125 m/s to push the 0.75 N the drag takes; removing that error needs integral action, since the 49 N a large enough P gain would ask for is far beyond the 4 N motors.
- A step response is described by rise time, overshoot, settling time and steady-state error, so different gains can be compared and requirements written before tuning; these numbers suit outputs that should follow their command, not a ramp or a fading push.
- Putting the controller into the robot's motion gives the error dynamics, so the PD heading loop behaves as a block on a spring (K_p) and a damper (K_d), and its swing can be predicted.
- A first-order loop closes 63 % of its gap per time constant, so one number, here 1 s, fixes its rise time (2.20 s) and settling time (3.91 s), and it can never overshoot.
- In a second-order loop the damping ratio alone sets the overshoot, because the natural frequency cancels from its formula, so overshoot requirements become damping

requirements.
- The natural frequency only stretches time, so a faster response needs a larger P gain, and the torque limit caps it at 2 rad/s for a 0.5 rad turn.
- The roots of the characteristic equation decide the kind of response: complex roots swing, a double root is critical damping, two real roots rise slowly, so where the roots sit tells the shape of the response.

We can now read a step response and choose the gains of a first-order or second-order loop for a requirement. One case remains: integral action, a lagging sensor or a time delay make the loop third order or higher, where these formulas fail and even stability is in question. Writing each loop as a transfer function and reading its poles and zeros handles them.

9. Sources

Built from

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- PX4 Autopilot, https://github.com/PX4/PX4-Autopilot: `src/modules/ mc_autotune_attitude_cor mc_autotune_attitude_control.cpp` (the rise time and damping index passed to the gain design), `src/modules/ mc_autotune_attitude_control/ mc_autotune_attitude_control_pa` (`MC_AT_RISE_TIME`, default 0.14 s), `src/lib/pid_design/pid_design.hpp` (“0 is critical damping, 1 is Butterworth”); read for the numbers quoted
- Autoware Universe, https://github.com/autowarefoundation/autoware_universe: `control/ autoware_pid_longitudinal_controller/ config/ autoware_pid_longitudinal_cont` and `src/pid_longitudinal_controller.cpp` (the overshoot emergency); read for the numbers quoted

10. Key terms

Terms taught in this Note come first; linked terms are recaps, taught in the Note the link opens.

Term	Meaning
Step input (G-2980)	A command that jumps at once from one steady value to another and then stays, such as a heading command going from 0 to 0.5 rad; it is the simplest test of a loop, because everything the output does afterwards comes from the loop itself.
Step response (G-2981)	The output of a system after a step input, such as the robot’s heading after its command jumps to 0.5 rad; its rise time, overshoot, settling time and steady-state error describe how well a loop follows a command.
Rise time (G-2982)	The time a step response takes to go from 10 % to 90 % of its final value (some books use 0 % to the first time it reaches 100 % for a response that overshoots), such as 2.20 s for the robot’s speed loop; it says how quickly a response rises.
Overshoot (step response) (G-2983)	How far a step response’s peak goes past its final value, as a percentage of the final value, such as 52.7 % when a 0.5 rad turn peaks at 0.763 rad; it says how far a response goes past its target.

Term	Meaning
Settling time (G-2984)	The time after which a step response stays within a band around its final value, usually 2 %, such as 2.92 s for the robot's critically damped turn; it says when a response has settled.
Steady-state error (G-2985)	The command minus the value a step response settles at, such as 0.125 m/s when a speed loop asked for 0.5 m/s settles at 0.375 m/s; a P loop keeps one whenever it must keep pushing against a load.
Error dynamics (G-2986)	The differential equation the error of a controlled system obeys, found by putting the controller's rule into the plant's equation of motion, such as 0.1 e" + 0.08 e' + 0.4 e = 0 for the heading under PD control; its solution predicts the step response from the gains.
First-order system (G-2987)	A system whose output follows a differential equation with only a first derivative, such as the robot's speed under P control; it closes the same fraction of its remaining gap in every time constant and can never overshoot.
Time constant (G-2988)	The time a first-order system takes to close 63 % of the gap to its final value, m/(K_v + b) = 1 s for the robot's speed loop; it fixes the rise time (about 2.2 time constants) and the 2 % settling time (about 4).
DC gain (G-2989)	The ratio of a stable system's final output to the size of a step input, such as 0.75 for the speed loop (0.375 m/s for a 0.5 m/s command); it says how much of a steady command the output ends up delivering.
Second-order system (G-2990)	A system whose output follows a differential equation with a second derivative, such as the robot's heading under PD control; written e" + $2\zeta\omega$_n e' + ω_n^2 e = 0, its response is set by two numbers, the natural frequency and the damping ratio.
Natural frequency (G-2991)	The rate, in rad/s, at which a second-order system would swing with no damping, $\omega_n = \sqrt{K_p/I}$, 2 rad/s for the heading loop; it sets how fast the response is, and raising it compresses the whole response in time.

Term	Meaning
Damping ratio (G-2992)	A second-order system's damping compared with the damping that just stops it from swinging, $\zeta = K_d/(2\sqrt{K_p I})$, 0.2 for D gain 0.08 and 1 for 0.4; it alone sets the shape of the response and its overshoot.
Underdamped (G-2994)	A second-order system with damping ratio below 1: its characteristic equation has two complex roots, so its step response swings past the target and fades, such as the 0.5 rad turn peaking at 0.763 rad with $\zeta = 0.2$.
Critically damped (G-2995)	A second-order system with damping ratio exactly 1: its characteristic equation has one double real root, so it gives the fastest step response that does not overshoot, such as the turn with K_p = K_d = 0.4 settling in 2.92 s.
Overdamped (G-2996)	A second-order system with damping ratio above 1: its characteristic equation has two different real roots, so its step response rises slowly without overshoot, held back by the slower root, such as a 7.44 s turn at $\zeta = 2$.
Characteristic equation (G-2993)	The polynomial equation found by trying a solution e^(st) in a linear differential equation, such as $s^2 + 0.8\ s + 4 = 0$ for the heading error; its roots say whether the solution fades, swings or grows.
Damped natural frequency (G-2997)	The rate, in rad/s, at which an underdamped second-order system actually swings, $\omega_d = \omega_n\sqrt{1-\zeta^2}$, 1.96 rad/s for ω_n = 2 and $\zeta = 0.2$; it is slower than the natural frequency because the damping drags on every swing, and it sets the peak time.
Peak time (G-2998)	The time an underdamped step response takes to reach its first and highest peak, half a damped swing, π/ω_d, such as 1.60 s for the first-try turn; the overshoot is read at that moment.
Heading θ (G-2284)	The angle from the world x-axis to the direction a robot faces, measured counter-clockwise; it decides which way the robot moves when it drives forward.

Term	Meaning
Proportional control (P control) (G-2958)	Feedback that sets the command in proportion to the control error, such as a torque of 0.2 N m per rad of heading error; it pulls the output towards the reference like a spring, but alone it swings when nothing damps the motion and leaves an error under a steady load.
PD control (G-2962)	Feedback with a proportional part and a derivative part, such as K_p e + K_d de/dt; it acts like a spring and a damper, so it can bring a frictionless joint or heading to its target without swinging.
Newton's second law for rotation (G-2957)	The net torque on a body equals its moment of inertia times its angular acceleration, tau = I alpha; it gives how fast a torque spins a body up, the turning form of F = ma.
Moment of inertia (G-2955)	How hard a body is to spin up or slow down about an axis: the sum of each small piece of mass times its distance from the axis squared, such as 0.1 kg m^2 for the robot about its vertical axis; it plays for turning the part mass plays for driving.
Newton's second law (G-2809)	The net force on a body equals its mass times its acceleration; it gives the force needed to change an object's speed, which the quasistatic assumption neglects.
Derivative (G-595)	The slope of a function at a point: how fast the output changes as the input changes there. It is zero at the lowest point of a smooth curve, which is how training finds a minimum.
Torque (G-2808)	The turning effect of a force about a point: the force times its distance from the point, measured square to the force's line; comparing torques decides whether a pushed object tips.
Proportional gain (G-2959)	The factor K_p by which a proportional controller multiplies the error, such as 0.2 N m per rad or 20 N per m/s; a larger gain gives a stronger correction, and for an undamped plant swings faster.

Term	Meaning
Derivative gain (G-2961)	The factor K_d by which derivative action multiplies the error's rate of change, such as 0.283 N m per rad/s; too little leaves a swing, too much makes the response sluggish.
Integral action (G-2963)	A part of the command in proportion to the error's running sum (its integral over time); the sum grows while any error remains, so it removes the error a steady load leaves, at the cost of overshoot and, if too large, instability.